

Attention Lecture 4

2026-08-24

Lecture 3

I'm a bit confused after that lecture. If i walk into a room and just look at nothing in particular, am I bottlenecking a particular feature, or am i simulatenously taking everything in?

You are:

- Simultaneously processing color, motion, but not their combination
- Extensively processing (through the bottleneck) one or a few objects at the location of your attention
- Doing additional processing that we don't understand through your non-selective pathway

Combining features within a dimension

[globalShape.html](#)

CHANGE BLINDNESS - Chapter 6

Points to “a grand illusion of visual experience”?

Blank-screen sandwiches: Lessons

- Grand illusion of visual experience?
- Peripheral vision
- Motion detectors
 - Blank screen
 - Gradual changes
- Memory

Find the change





Change blindness

Be able to explain why it inspired the “grand illusion of visual experience” theory





It feels like we see everything at once

- But, change blindness!
- So, we're wrong that we see everything simultaneously.
- There's a "grand illusion of visual experience"
- It's like the refrigerator light illusion

Argument for “the grand illusion”

- **Premise 1.** If you visually experience most of a scene’s objects, you will notice major changes to them.
- **Premise 2.** You don’t detect major changes to most objects in a scene (in the blank-screen sandwich movies).
- **Conclusion.** Therefore, you *don’t* visually experience most objects in a scene.

If A, then B.

B is not true.

Therefore, A is not true.

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Answer a question about the “grand illusion of visual experience” argument.

Argument for “the grand illusion”

Premise 1. If you visually experience most of a scene’s objects, you will notice major changes to them.

Premise 2. You don’t detect major changes to most objects in a scene (in the blank-screen sandwich movies).

Conclusion. Therefore, you don’t visually experience most objects in a scene.

Needed to detect a change

1. Internal representations of the object before and after the change.
 2. A process that compares the before representation to the after.
 3. A process that calls attention to, or brings into conscious awareness, the comparison results.
- **Premise 1.** If you visually experience most of a scene's objects, you will notice major changes to them.
 - Does having visual experience of all the objects in a scene mean you also have 1 and 2?

The processes for detecting a change

1. Before / after

2. Comparison

Premise 1. If you visually experience most of a scene's objects, you will notice major changes to them.

- Premise 1 is questionable
- You might be able to have visual experience of the objects without you having 1 and/or 2

Argument for the grand illusion

Premise 1. If you visually experience most of a scene's objects, you will notice major changes to them.

Premise 2. You don't detect major changes to most objects in a scene (in the blank-screen sandwich movies).

Conclusion. Therefore, you don't visually experience most objects in a scene.

- Premise 1 was part of the argument for the grand illusion
- Premise 1 is highly questionable
- The argument for the “grand illusion” is not good!

Reconciling phenomenal experience with experiment findings

1. Introspection says we experience the entire visual field simultaneously.
2. Results of experiments find knowledge of what's in the visual field is very limited, to only several objects ¹
1. How do we reconcile 1 and 2?
1. If you do things to prevent attention wandering around and building up the scene representation over time

Feels like we see everything at once

↑ How can we reconcile this with the bottleneck?

1. Attention shifts so quickly we don't notice it

- The refrigerator light illusion.

2. Something's missing from our theories?

Resolving the puzzle

Ned Block:

We **do** visually experience the whole visual field simultaneously. Allow me to explain how that's compatible with the results of the experiments... [writes 550-page book]

“Grand illusion of visual experience” people:

Block, you're so wrong. But we know you strongly believe you do experience the whole visual field simultaneously. We want to explain why you believe it so strongly, so we'll posit a grand illusion of visual experience throughout the visual field. Maybe it's kinda like the refrigerator light illusion.

Ned Block

The Border
Between
Seeing
and Thinking

Argument for the grand illusion

Premise 1. If you visually experience most of a scene's objects, you will notice major changes to them.

Premise 2. You don't detect major changes to most objects in a scene (in the blank-screen sandwich movies).

Conclusion. Therefore, you don't visually experience most objects in a scene.

The argument relies on having visual experience of the whole visual field entailing that you'd detect major changes to the visual field.¹ But based on our analysis of what change detection involves, it may not entail that.

So that argument for the "grand illusion" is not very good.

1. if you don't know what "entail" means, please look it up :)

It feels like we see everything at once

- Explanation: “phenomenal content overflows cognitive access.” - Ned Block
- An alternative:
 - We experience less than we think
 - A “grand illusion of visual experience” (6.1)

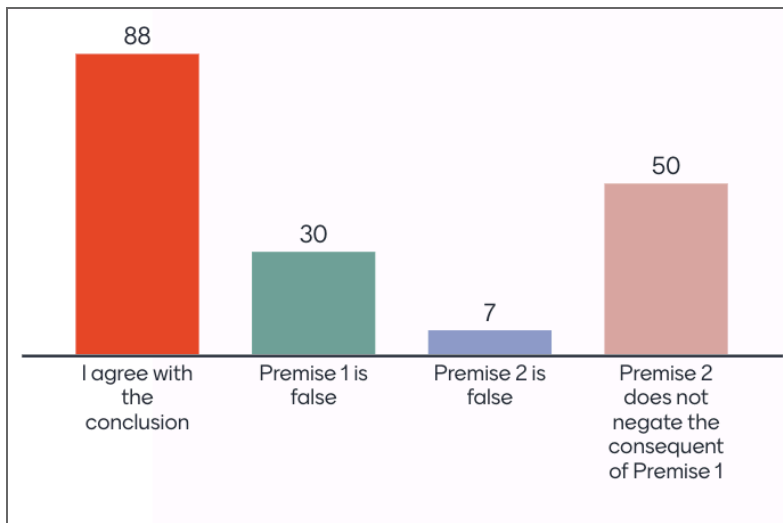


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Review of the argument, now in essay format in case it helps

The “grand illusion” argument: if we really did experience everything in the scene, we would quickly detect changes to the objects in the scene. So, because we don’t quickly detect many changes to the scene (change blindness), we must *not* really experience everything.

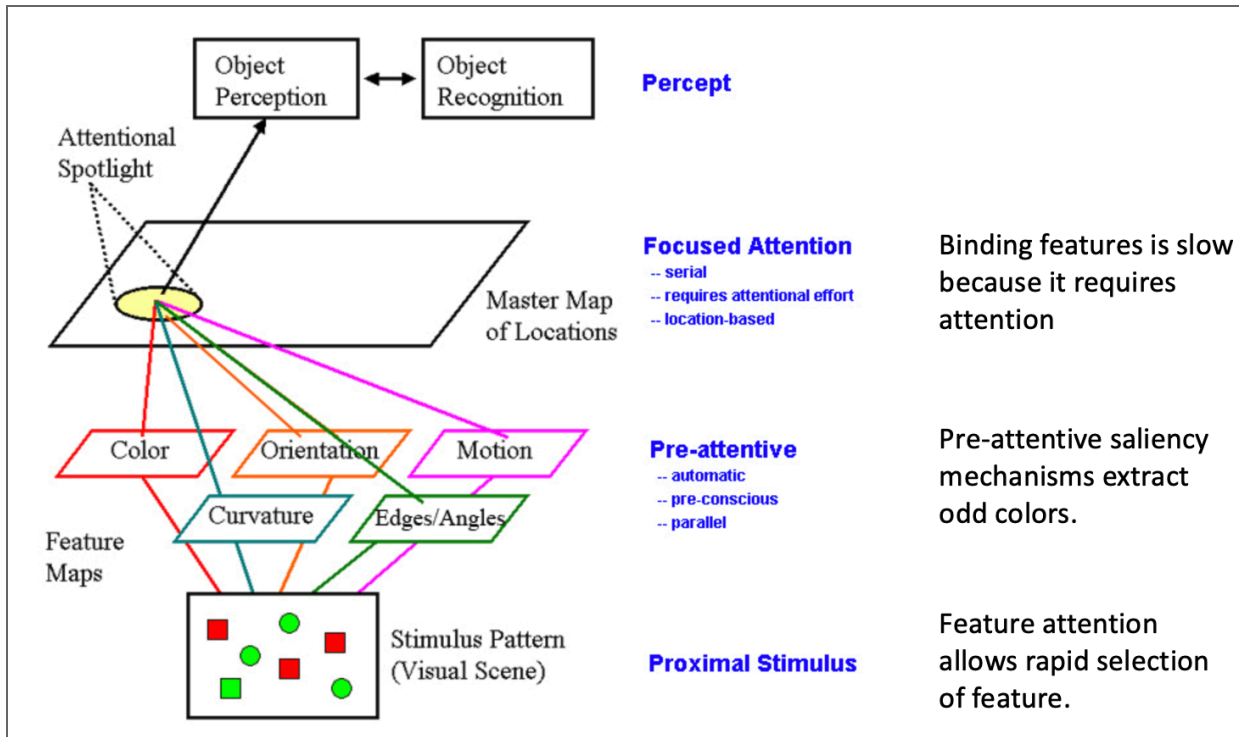
After hearing the argument, about half of students agreed that our feeling of experiencing everything must be only an illusion.



But the grand illusion argument is actually *not* a great argument; the first part that says that if we really did experience everything, we would quickly detect changes. Maybe we really do experience everything but we don’t have a comparison process that’s always comparing everything. Or maybe we really do experience everything but we don’t have a representation of the scene a short time ago, so we can’t do the comparison. In other words, the idea that “if we really did experience everything, we would quickly detect changes” may be false.

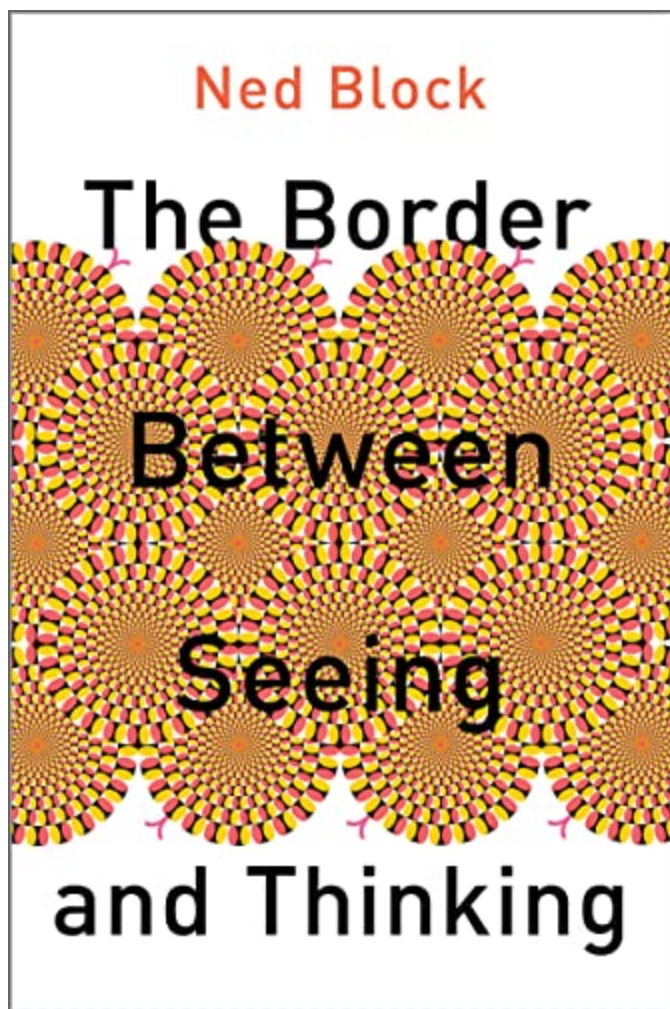
Explaining blank screen sandwiches

- Focused attention is needed to detect change



Change blindness -> “the grand illusion”

- You can still believe there is a “grand illusion” of visual experience, but change blindness is not a good argument for it.
- There is *other* evidence that we only have access to a few visual objects



For more, see Overgaard (2018).

Blank-screen sandwiches

- Peripheral vision
- Motion detectors
- Memory



Peripheral vision

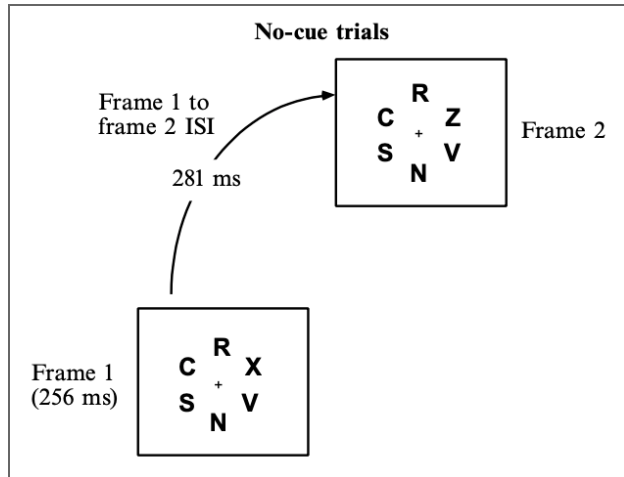
Good experimenters ensure that the object is visible in the periphery.

- If the object is not visible in the periphery, need *overt* attention to the location, not just covert attention
- Unsurprising that this can't be done at all locations at once!

Peripheral vision



Poor peripheral vision



- In experiments, stimuli are placed carefully so they are visible in the periphery



Becker, Anstis, and Pashler (2000)

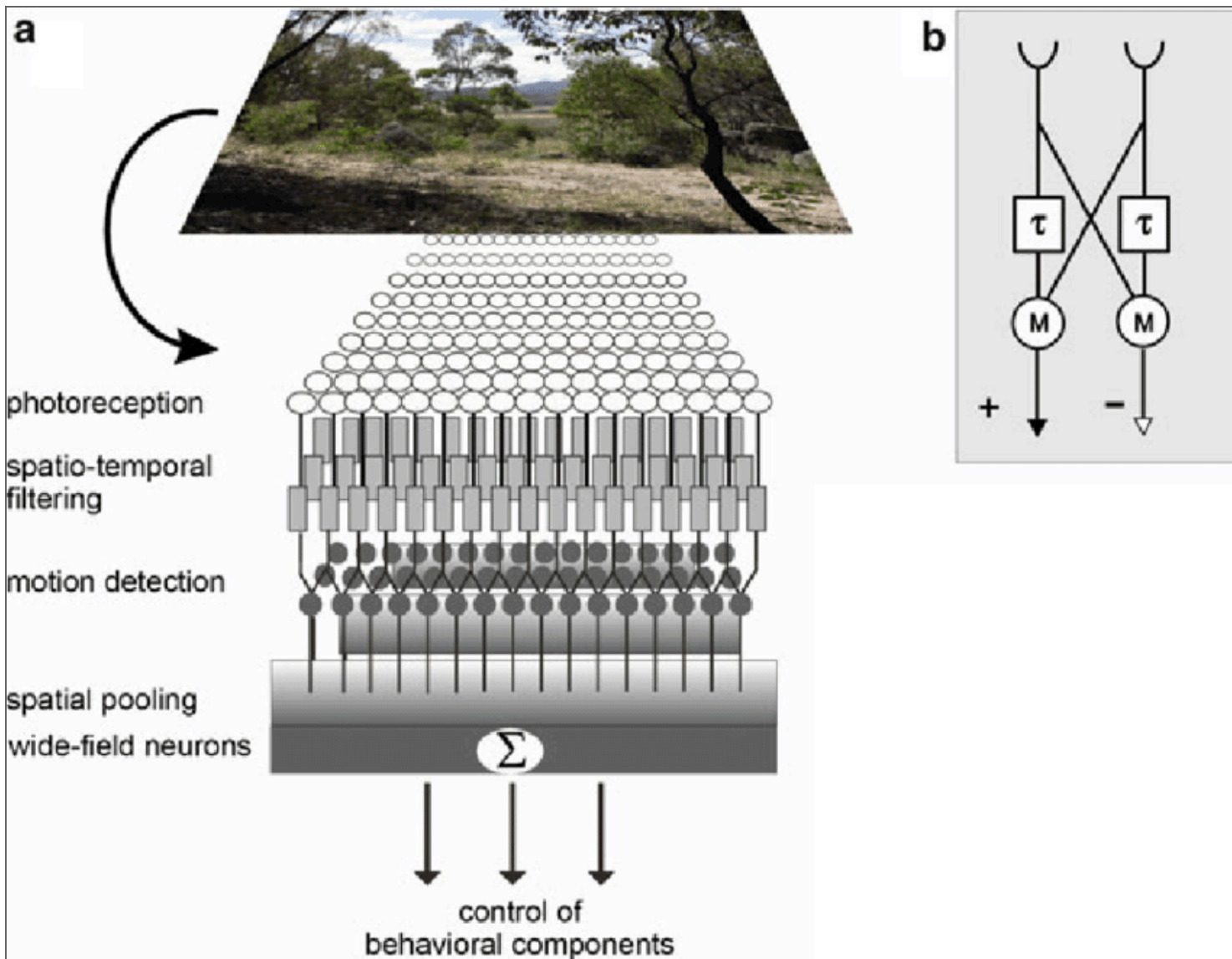
Once you find it, it's hard to un-see it



- Memories guide attention quickly
- This helps us in countless situations, including on the road

No blank frame







- Motion detectors respond to flicker
- If there's motion everywhere, there might as well be motion nowhere
- Odd motion detector response no longer makes change location salient



- “Mud splashes” almost as good as blank screen
- Motion detectors misdirect attention

What made the mud splashes more salient than the white road lines?

- They are bigger and thus stimulate the motion detectors more strongly
- They are more different to their surround (parallel bottom-up attention processing)

Gradual changes

Video player configuration error

Error 153

Watch video on YouTube

Youtube

Gradual changes

Why are gradual changes hard to find?

- They don't stimulate motion detectors

Blank-screen sandwiches: Lessons

- Grand illusion of visual experience?
- Peripheral vision
- Motion detectors
 - Blank screen
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Tutorial Week 5

Implications for driving

Change blindness in continuous movie

Video player configuration error

Error 153

Watch video on YouTube

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Video player configuration error

Error 153

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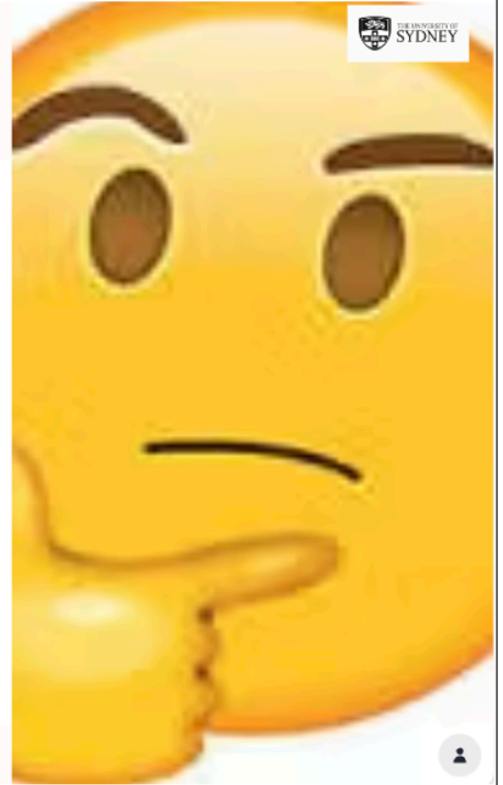
Why do people miss many of the changes?

- Did you notice that this slide has no background image? 😊

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Thing(s) I didn't understand

Waiting for responses ...



Didn't understand:

The diagrammed model in the Shapes slides

Come up and see me

What were the answers to the lecture quiz questions? :O

Sorry about that, it was "it stimulates motion/flicker detectors in every location..

Feels like the start of the recording was chopped off, could you please check that it's recording before you commence?

Sorry about that.

what made the mud splashes more salient than the road lines

They are bigger and stimulate the motion detectors more strongly. They may also benefit from other salience processes because they are more different from their surroundings than the road lines.

can you explain motion detectors a bit more?

There's more in PSYC3014. For this unit you just need to know they are stimulated by flicker as well as motion and odd motion/flicker relative to neighboring parts of scene attract attention. Or come talk to me.

Can you summarize the first point about the two conditions and conclusion?

Come up and see me and we can talk about the summary slides for that.

I wanted to say that I have been reading your mini text, and I am finding it really helpful to support my understanding.

I'm so glad to hear that!

We learnt that memory helps us attend to changes in the world around us - is this linked to the phenomena where, for example, your friend gets a rare car, you start to notice those cars more?

Yes! And that phenomenon is discussed in Chapter 13

We learnt that memory helps us

PSYC2016: Attention

Speaker notes